



QUALITY SYSTEM

FORM

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TITLE:

LABORATORY OOS INVESTIGATION FORM

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OOS Number: OOS- 262074

Phase I

Section A – Test Details

Initiator:	Alex Saravia (Written by: Qiyue Chen)	Test Date:	31-Aug-2026	Date Initiated:	07-Sep-2026
Date of Incident:	07-Sep-2026	Test Name:	USP <71> / EP 2.6.1 Sterility Test	Sample ID:	ETX-260806-0702 (BUD-T30)
Name of Analyst who Performed the Test:	Prepper: Andrew Carrillo (AC)	Sample / Active Name	TB-500 10mg/ml	Lot #:	24JUL2026-059
	Processor: Alex Saravia (ES)	Dosage Form	Liquid		
	Reading Analyst: Andrew Carrillo (AC)				
Description of Incident:	On 07Sep26, sample ETX-260806-0702 (BUD-T30) was found positive for viable microorganisms after USP <71> / EP 2.6.1 sterility testing.		☒ Client Care notified of OOS ☒ Sterility and Endotoxin test related OOS; Client must be notified within 24 hours of OOS		
	SOP / Test Method # 2.600.008	Effective Date: 26Aug25	SOP / Test Method Rev: 18	Limits/Specification: Pass/Fail	
Manager notified	Manager/designee name: Kathan Parikh	Date: 07-Sep-2026	Quality Assurance Department notified:	Yes ☒ No ☐	

Section B – Investigation and Root Cause Analysis

Personnel Discrepancy		Comments
Analyst interviewed?	☒ Yes ☒ No ☐ N/A	Yes, analysts Andrew Carrillo and Alex Saravia were interviewed comprehensively.
Correct samples analyzed?	☒ Yes ☒ No ☐ N/A	Yes, sample ID: ETX-260806-0702
Correct SOP / Test Method used/followed?	☒ Yes ☒ No ☐ N/A	Yes, as per SOP 2.600.008
Correct sample collection/technique?	☒ Yes ☒ No ☐ N/A	Yes, as per SOP 2.600.008
Are the Test Result Records Attached & Calculations Verified and Correct?	☒ Yes ☒ No ☐ N/A	Yes, Information is available in EagleTrax under ETX-260806-0702
Is the analyst qualified to perform the test?	☒ Yes ☐ No ☐ N/A	Yes, all analysts are trained and qualified by quality to perform the test.
Is formula worksheet provided and verified?	☐ Yes ☐ No ☒ N/A	Not Applicable
Does the processing method/method set conform to the required specification?	☒ Yes ☐ No ☒ N/A	Not Applicable
Was the sample stored appropriately?	☒ Yes ☐ No ☐ N/A	Yes, the sample was stored refrigerated as per client's instructions. Information is available in EagleTrax Sample Location History under ETX-260806-0702
Material Discrepancy		Lot Number
Is the integrity of the sample(s) questionable?	☐ Yes ☐ No ☐ N/A	
Was the sample transported appropriately?	☒ Yes ☒ No ☐ N/A	
Are the other test results from this run under investigation?	☐ Yes ☒ No ☐ N/A	



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Appropriate/correct glassware used?	☐ Yes ☐ No ☐ N/A		
Are reagents stored in proper condition and not expired?	☒ Yes ☐ No ☐ N/A	Process Consumables: FTM 100ML: 2567320 TSB 100mL: 2548010 Fluid D: 685561	Process Consumables: FTM: 12/16/2026 TSB: 04/19/2027 Fluid D: 10/31/2026
Standard Used	Not Applicable	Not Applicable	Not Applicable
Solvent Used	Not Applicable	Not Applicable	Not Applicable
Is the standard the source of the root cause?	☐ Yes ☐ No ☐ N/A		
Has the analyst been instructed to keep the original test solution for possible retesting?	☐ Yes ☐ No ☐ N/A		
Equipment Discrepancy		Equipment ID	Calibration Due Date
Was the instrument set up correctly?	☒ Yes ☐ No ☐ N/A	ISO 5 BSC E001314	Dec 2026
Is the clean room facility certified?	☒ Yes ☐ No ☐ N/A	CR115 (E001737)	Jun 2026
System Performance Check (if applicable)?	☐ Yes ☐ No ☐ N/A	E001314	Not Applicable
System Suitability RSD < 2% (PDA)?	☐ Yes ☐ No ☐ N/A	Not Applicable	Not Applicable
System Suitability RSD < 10% (QDA, CAD)?	☐ Yes ☐ No ☐ N/A	Not Applicable	Not Applicable
Do the environmental monitoring results conform to required standards?	☒ Yes ☐ No ☒ N/A	See Phase I Summary	See Phase I Summary
Do the controls (i.e., positive, negative, etc.) conform to the required standards?	☒ Yes ☐ No ☒ N/A	See Phase I Summary	See Phase I Summary
Is equipment/instrument functional and calibrated/qualified/certified/PM?	☒ Yes ☐ No ☒ N/A	E001356 (Monitored by Sensor E001450) E001357 (Monitored by Sensor E001449)	Jul 2026 / Feb 2027 Jul 2026 / Feb 2027



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Are chromatogram/spectra/raw data correct?

☐ Yes ☐ No ☐ N/A

Not Applicable

Not Applicable

Section C – Evaluation of Laboratory Testing/Hypothesis Testing (if applicable)

☒ N/A☐ Re-injection of sample vial☐ Re-injection of original standard vial☐ Re-vialing of original working standard solution☐ Re-vialing of original working sample solution☐ Re-dilution of a working sample solution from the original stock☐ Re-dilution of a working standard solution from the original stock standard☐ Other: N/A QC 18Sep26

Section D – Phase I Summary

Phase I Summary:

All analysts involved in the prepping, processing, and reading of the sample – Andrew Carrillo (AC) and Alex Saravia (ES) were interviewed comprehensively. Their answers are recorded throughout this document.

Upon arrival, the sample was stored in accordance with the Client ' s instructions under refrigeration. Analyst Alex Saravia verified the integrity of the sample throughout both the preparation and processing stages. No leaks or turbidity were observed at any point, verifying the integrity of the sample.

All reagents and supplies mentioned in the material section above were stored according to the suppliers ' recommendations, and their integrity was visually verified before utilization. Fluid Thioglycollate Medium (FTM 100ML, Lot: 2567320, Exp: 12/16/2026), Trypticase Soy Broth (TSB 100mL, Lot: 2548010, Exp: 04/19/2027), and Fluid D (Lot: 685561, Exp: 10/31/2026) had valid expiration dates. The functionality of all equipment was confirmed by reviewing data generated by our comprehensive in-house continuous monitoring system.

The cleanroom used for processing procedures (Suite 115) comprises three interconnected sections: the innermost ISO 7 cleanroom (115B), which connects to the middle ISO 7 buffer room (115A), and then to the outermost ISO 8 anteroom (115). A positive air pressure system is maintained throughout the suite to ensure controlled, unidirectional airflow from 115B through 115A and into 115.

The ISO 5 BSC E001314, located in the innermost ISO 7 room (115B), was used for sample processing steps. It was thoroughly cleaned and disinfected prior to each procedure in accordance with SOP 2.600.018 (Cleaning and Disinfecting Procedure for Microbiology). Additionally, BSC E001314 was certified and approved by both the Engineering and Quality Assurance teams. Sample processing was conducted in the ISO 5 BSC E001314 in 115B by Alex Saravia on 31Aug26 as per SOP 2.600.008 (USP <71> / EP 2.6.1 Sterility Test).

On 28Aug26, the sample containers for ETX-260806-0702 were received from the Sample Submissions team and brought into the Sterile Microbiology lab. Upon receipt, each sample container was sprayed with an acidified bleach disinfectant, placed into pre-disinfected bins, and allowed a 10-minute contact time. Following initial disinfection, the bins were transferred to the ISO 8 anteroom (115), where the containers underwent a second disinfection using acidified bleach, again with a 10-minute contact time. Subsequently, the containers were moved into the ISO 7 cleanroom 115B. Inside this cleanroom, the processing analyst, Alex Saravia, performed a final disinfection step, allowing an additional 10-minute contact time. Once fully disinfected, the containers were transferred into the ISO 5 BSC E001314.

Once transferred into the ISO 5 BSC, the containers were placed on the disinfected working surface of BSC E001314 and aseptically processed in accordance with SOP 2.600.008 (USP <71> / EP 2.6.1 Sterility Test) utilizing membrane filtration under method suitability ETX-260623-0160. Inside the BSC, each container was aseptically opened, ensuring minimal exposure to the environment. The contents of 10 vials (20 mL per media) were filtered directly through sterile canisters on a vacuum manifold. Following filtration, canisters were rinsed with sterile Fluid D and filled with TSB and FTM media. Media bottles were transferred into incubators E001356 (monitored by sensor E001450) and E001357 (monitored by sensor E001449) to begin incubation. Visual inspections were performed periodically by the analyst over the 14-day incubation period to detect any macroscopic evidence of microbial growth.

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Phase I Summary (continued):

During periodic visual inspections on 07Sep26 (Day 7 of incubation), macroscopic evidence of microbial growth was observed by reading analyst Andrew Carrillo in 1 x 100 mL TSB media bottle, while FTM media bottles showed no microbial growth. The positive TSB bottle for ETX-260806-0702 was submitted for Differential Staining and Microbial Identification under ETX-260907-0017, where the organism was definitively identified as *Paenibacillus lautus* Gram (+) rods.

The culture media utilized were within their expiry period. The negative control canisters for the membrane filtration method were handled, processed, and incubated in a manner identical to that of actual samples. No microbial growth was observed in the corresponding negative controls.

A review of the Environmental Monitoring (EM) results for the relevant testing period showed no microbial growth on daily personnel monitoring plates (left and right glove touches), daily settling plates, or ISO 5 BSC E001314 surface sampling plates across all bracketing dates (28Aug26, 31Aug26, and 01Sep26). Weekly cleanroom surface sampling for Cleanroom 115 (E001737), performed on 04Sep26, likewise demonstrated no microbial growth. Weekly active air sampling on 04Sep26 showed 1 CFU (ETX-260914-0487, pending identification) in the ISO 8 anteroom (114). However, all sample processing activities were performed within the validated ISO 5 BSC E001314 located in innermost ISO 7 cleanroom 115B. The sample containers were transferred inside closed, disinfected bins, and did not come into contact with ambient ISO 8 air. Furthermore, the complete absence of microbial recovery across all daily analyst glove touch plates, critical ISO 5 BSC work surfaces, and settling plates confirms that no contamination transfer pathway existed between the lower-classified anteroom environment and the ISO 5 processing zone.

The analysts confirmed full compliance with cleaning procedures as outlined in SOPs 2.600.018 (Cleaning and Disinfecting Procedure for Microbiology) and 2.600.008 (USP <71> / EP 2.6.1 Sterility Test). Monthly cleaning and disinfection of the outermost ISO 8 Anteroom, the middle ISO 7 Buffer room, the innermost ISO 7 cleanroom, and its containing ISO 5 Biosafety Cabinets for 115 was performed on 30Aug26, as per SOP 2.600.018. During both cleaning cycles, all H2O2 indicators passed, confirming efficient monthly decontamination.

Analyzing a 6-month sample history for Hillstone Pharmacy (E17047), this specific analyte "TB-500 10mg/ml" has had no prior failures using USP <71> / EP 2.6.1 Sterility Test during this period. Furthermore, the previous T0 time point for this stability lot (24JUL2026-059) yielded valid passing sterility results.

To assess the potential for sample-to-sample contamination contributing to the positive results, a comprehensive review was conducted of all samples processed on the same day (31Aug26). Sample ETX-260806-0702 was the 4th sample processed in the batch. All other samples processed by the analyst in the batch on that day tested negative. These findings confirm that cross-contamination between samples is highly unlikely.

Based on the observations outlined above, it is unlikely that the failing results were due to reagents, supplies, the cleanroom environment, the process, or analyst involvement. Consequently, the possibility of laboratory error contributing to this failure is minimal and the original result is deemed to be valid.



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Phase I Summary (continued):

Reviewer comment(s): Satisfactory investigation. The failure is valid and not attributable to laboratory error.



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Has root cause been identified as a possible laboratory error?

Note: Root cause must be conclusive, verifiable, and well-documented.☐ Yes☐ If retesting is **required**, complete Phase II of the OOS (initiate Form 3.100.019.F02).☐ No, the OOS result is valid.

Reviewer comment(s):

Prepared by (Print):

Alex Saravia (Written by: Qiyue Chen)

Signature/Date:

Lab Manager/Designee (Print):

Robin Seymour

Signature/Date:

Section E – Quality Disposition

Can investigation be closed?

☐ Yes☐ Original result is valid.☐ No, initiate Form 3.100.019.F02.

Approved by QA Representative (Print):

Signature/Date:

Table 1: Information for ETX-260806-0702 (BUD-T30) under investigation

Reading Analyst	Day of Incubation	Sample ID	Related Microbial ID	Media with microbial growth	Microbial Identification
Andrew Carrillo	7	ETX-260806-0702 (BUD-T30)	ETX-260907-0017	1 x 100 mL TSB	Paenibacillus lautus Gram (+) rods

Table 2: Environmental Monitoring for Analyst & Cleanroom

Environmental Monitoring (EM) Sampling Site	Frequency	Date (DDMM YYYY)	Analyst (Initials)	Day /Week(s)	Observation	Environmental Monitoring Plate ETX ID	Microbial ID	Notes
Personnel EM Bracketing 31Aug2026								
Personal (Left Touch and Right Touch)	Daily	28 Aug 2026	ES	Date Before Testing	No growth	N/A	N/A	None
Personal (Left Touch and Right Touch)	Daily	31 Aug 2026	ES	Date of Testing	No growth	N/A	N/A	None
Personal (Left Touch and Right Touch)	Daily	01 Sep 2026	ES	Date After Testing	No growth	N/A	N/A	None
Biological Safety Cabinet EM Bracketing Biological Safety Cabinet (BSC) E001314								
Surface Sampling of ISO 5 (4 locations)	Daily	28 Aug 2026	ES	Date Before Testing	No growth	N/A	N/A	None
Surface Sampling of ISO 5 (4 locations)	Daily	31 Aug 2026	ES	Date of Testing	No growth	N/A	N/A	None
Surface Sampling of ISO 5 (4 locations)	Daily	01 Sep 2026	ES	Date After Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	28 Aug 2026	ES	Date Before Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	31 Aug 2026	ES	Date of Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	01 Sep 2026	ES	Date After Testing	No growth	N/A	N/A	None
Weekly Active Air Sampling for CR 114								
Active Air Sampling of Cleanrooms	Weekly	04 Sep 2026	SMO	Week of Testing	1 CFU (ISO8)	ETX-260914-0487	Pending	None
Surface Sampling of Anteroom and Cleanroom for CR 115 (E001737)								
Surface Sampling of Cleanrooms	Weekly	04 Sep 2026	SMO	Week of Testing	No growth	N/A	N/A	None